

Project Overview

- Modern soccer generates large volume of data. But translating these statistics into clear insights remains challenging.
- Using 2025 Wyscout data, we analyzed high-dimensional team and player statistics to identify patterns in team performance, player contributions, and match outcomes.
- Our goal was to transform complex soccer data into a practical evaluation framework that can support training priorities, player assessment, and match preparation.

Player Analysis

- How central is each player within their own team?

We combine availability, team action share, and role-adjusted quality into an interpretable player importance score.

$$Score_i = 100(\lambda_V V_i + \lambda_T T_i + \lambda_R R_i)$$

where i denotes a player, $t(i)$ is the player's team, and $g(i)$ is the player's position group.

- Availability

$$V_i = \sqrt{\frac{M_i}{A_{t(i)}}}$$

where M_i is player i 's total minutes and $A_{t(i)}$ is the estimated available playing time for one player on team $t(i)$.

- Team Action Share

$$T_i = PctRank_{t(i)} \left(\frac{1}{|K|} \sum_{k \in K} \frac{x_{ik}}{\sum_{j \in t(i)} x_{jk}} \right)$$

where x_{ik} is player i 's total for action k , including shots, passes, progressive actions, recoveries, and defensive actions, etc.

- Role-Adjusted Quality

$$R_i = \frac{1}{|C_{g(i)}|} \sum_{c \in C_{g(i)}} PctRank_{g(i)}(q_{ic})$$

$$r_{ik} = \frac{90x_{ik}}{M_i} \quad z_{ik} = \frac{d_k r_{ik} - \mu_{g(i),k}}{\sigma_{g(i),k}} \quad q_{ic} = \frac{1}{|F_c|} \sum_{k \in F_c} z_{ik}$$

where r_{ik} is the per-90 rate, and z_{ik} is its direction-adjusted z-score of the same position group. F_c denotes the metrics in component c , $C_{g(i)}$ denotes the components used for position group $g(i)$, and $PctRank_{g(i)}$ ranks players within that group.

Team Analysis

- Which metrics are strongly associated with goal difference?
- ElasticNet Feature Selection

We use ElasticNet to select a sparse set of informative predictors from highly correlated and high dimensional match statistics. The response variable is match-level goal difference:

$$y_i = Goals_i - Conceded Goals_i$$

After standardizing and dropping highly correlated predictors, we estimates

$$\hat{\beta}(\alpha, \rho) = \arg \min_{\beta} \frac{1}{2n} \|y - X\beta\|_2^2 + \alpha \left(\rho \| \beta \|_1 + \frac{1-\rho}{2} \| \beta \|_2^2 \right).$$

Here, α controls the overall penalty strength, while ρ balances L_1 and L_2 normalization. ElasticNet retains variables with nonzero coefficients. Then we refit OLS on selected feature set S .

- Two Measures of Feature Contribution

Partial R^2 : Unique Contribution ; LMG share R^2 : Shared Contribution

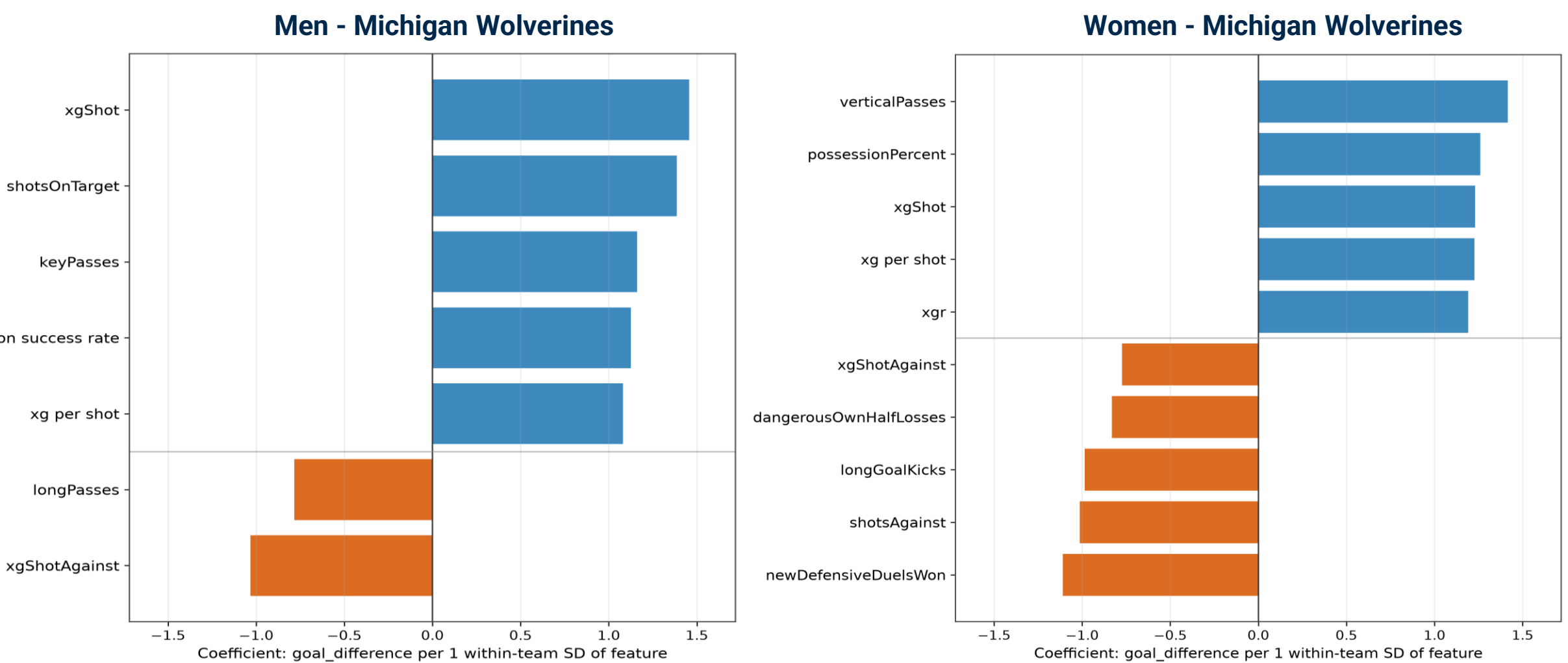
(a) Selected Significant Associations ($p < .10$)					
Men			Women		
Metric	$\hat{\beta}$	p	Metric	$\hat{\beta}$	p
Opponent xG	−0.772	< .001	GK saves	+0.399	< .001
Shots on target	+0.543	.001	Key-pass completion	+0.316	.003
Home advantage	+0.326	.002	Shots against	−0.776	.003
Team xG	+0.584	.005	Opponent strength	−0.226	.005
Missed balls	−0.257	.016	Opponent xG	−0.463	.009

(b) Feature Contribution to Explained Variation				
Method	Men		Women	
	Metric	Contribution	Metric	Contribution
Partial R^2	Opponent xG	7.6%	Key-pass completion	2.1%
	Home advantage	2.1%	GK saves	1.7%
	Shots on target	1.8%	Shots against	1.4%
LMG share	Shots on target	19.5%	xG ratio	8.0%
	Opponent xG	15.3%	Opponent xG	7.8%
	Team xG	14.8%	Shots on target	7.3%

- How do performance signals differ across teams?

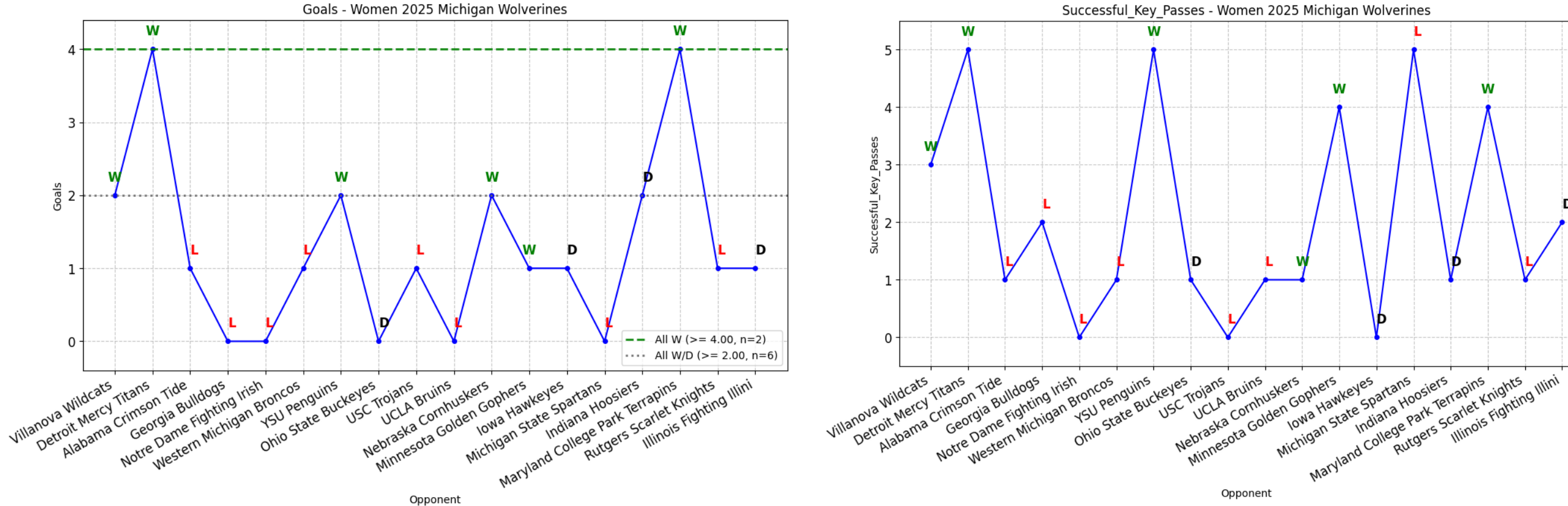
- Team-Specific Performance Profiles

For each team, we fit separate OLS models using team standardized metrics retained by ElasticNet. We display up to five positive and five negative associations with $p_{\text{value}} < .10$.



Match Outcome: Distribution-Based WDL

- How team-specific performance levels are associated with winning, avoiding defeat, or losing?
- WDL Plot with Threshold

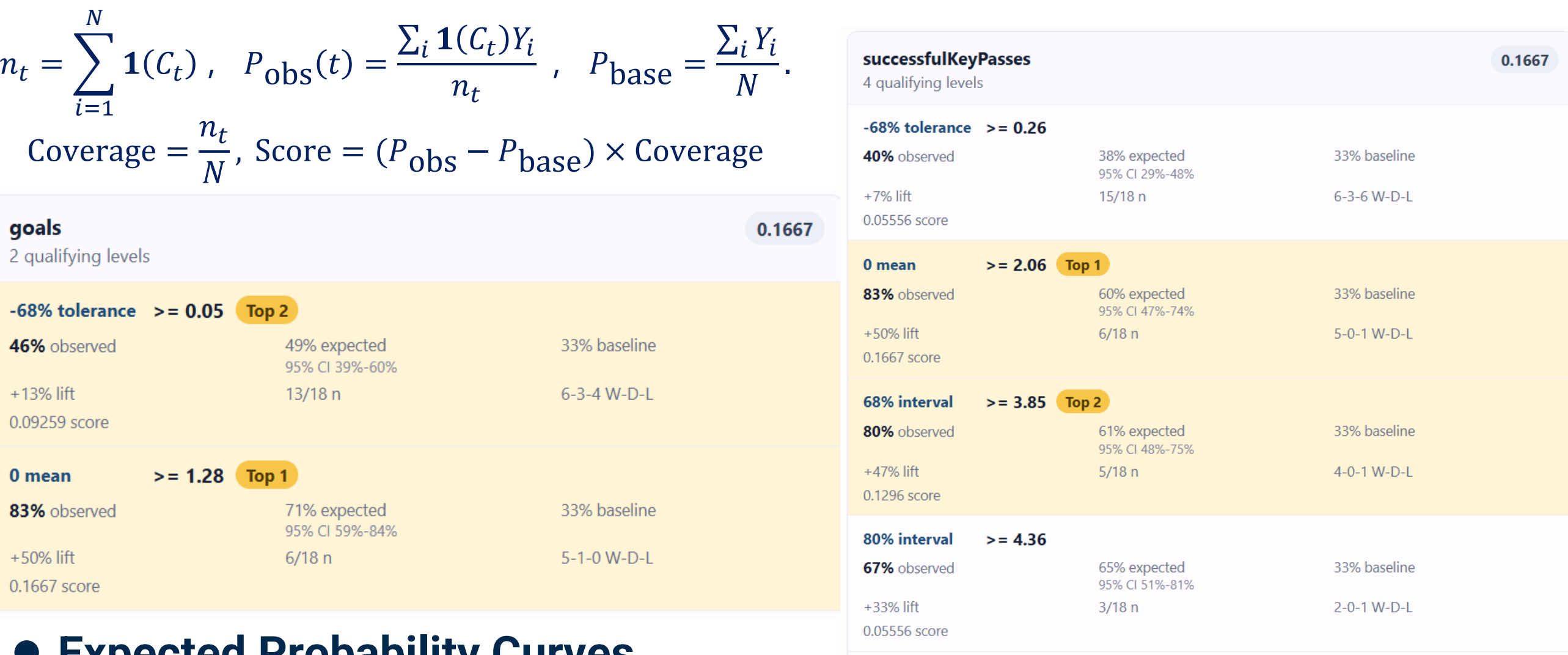


For each team and performance metric, we approximate match-level values by:

$$X \sim N(\bar{x}, s^2)$$

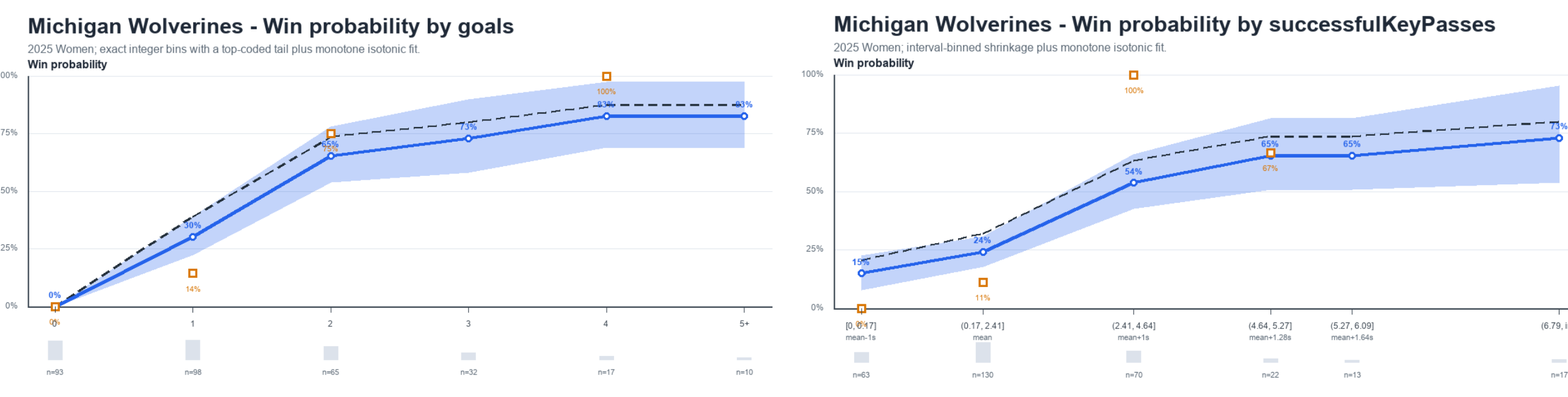
Candidate thresholds include the team mean and the bounds of the central

68%, 80%, 90%, and 95% intervals: $t_c^{\pm} = \bar{x} \pm \Phi^{-1} \left(\frac{1+c}{2} \right) s$



- Expected Probability Curves

Expected probability curves show how a team's chances of winning or avoiding defeat change with a performance metric.



Next Step

- Integrate the new WDL dashboard into the UM Soccer analytics website.
- Update pregame reports with team-specific performance insights to support match preparation and training priorities.

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- Students currently on the project.